



MONASH
University

Air Quality and Respiratory Health: An Analysis on Healthcare System Loading

Somya Verma

BAEco(Hons), MBAt

Daniel Loza

AMusA, BBus, BCom(Hons)

Yunhui Fan

BSc(Hons), MBAt

Yixiang Xu

BManSc, MBAt

Report for
6it Consultants

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MONASH
BUSINESS
SCHOOL

**Department of
Econometrics &
Business Statistics**

☎ (03) 9905 2478
✉ BusEco-Econometrics@monash.edu

ABN: 12 377 614 012



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1 Executive summary

2 Introduction

Question to investigate: **Impact on Hospital Intakes in top 5 most air-polluted cities of world from 2020-2025**

3 Methodology

The analysis was conducted using R and several libraries from the tidyverse ecosystem for data cleaning, transformation, and visualisation. Additional packages including lubridate, ggplot2, kableExtra, and patchwork were used for date handling, plotting, and professional table formatting.

The dataset contains daily air quality and healthcare observations from multiple cities between 2020 and 2025. Key variables include Air Quality Index (AQI), PM2.5, PM10, NO2, O3, temperature, humidity, and hospital admissions. The dataset was used to examine whether increasing pollution levels were associated with higher hospital intake rates in heavily polluted cities.

[1] "air_quality.csv"

To identify the most polluted cities, the average AQI for each city was calculated using grouped summary statistics. Cities were then ranked according to their average AQI, and the top five most polluted cities were selected for further analysis.

Daily observations were aggregated into monthly and yearly averages to reduce short-term fluctuations and improve trend interpretation. Average AQI, PM2.5 concentration, and hospital admissions were

calculated for each city and year. These summaries were presented using formatted tables and visualisations.

Table 1: Average AQI by City and Year (2020-2025)

City	2020	2021	2022	2023	2024	2025
Beijing	272.85	262.08	262.86	243.69	274.54	289.64
Cairo	240.61	234.00	317.00	185.25	417.60	193.14
Delhi	244.30	240.37	251.01	239.34	258.94	230.38
Los Angeles	254.17	246.21	215.96	260.52	241.39	243.10
São Paulo	392.00	165.35	113.92	196.92	181.08	190.08

A multiple linear regression model was used to examine the relationship between air pollution and hospital admissions. Hospital admissions were treated as the dependent variable, while AQI, PM2.5, PM10, NO2, and O3 were included as predictor variables.

Call:

```
lm(formula = hospital_admissions ~ aqi + pm2_5 + pm10 + no2 +
    o3, data = airquality_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.845	-2.307	-0.108	2.195	16.989

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.626e+00	6.898e-02	67.064	<2e-16 ***
aqi	-3.942e-05	7.952e-05	-0.496	0.620
pm2_5	9.870e-02	7.780e-04	126.867	<2e-16 ***
pm10	1.086e-04	5.804e-04	0.187	0.852
no2	-2.419e-04	1.153e-03	-0.210	0.834
o3	-8.434e-04	9.569e-04	-0.881	0.378

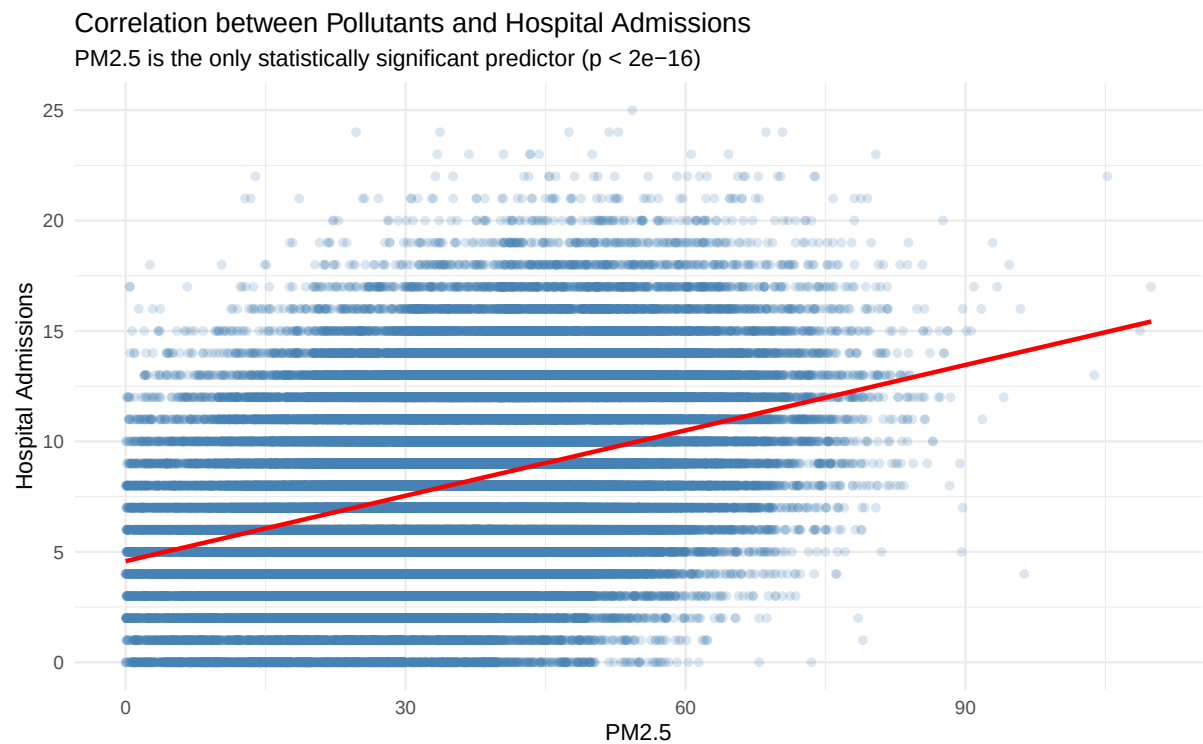
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.418 on 88483 degrees of freedom

Multiple R-squared: 0.1539, Adjusted R-squared: 0.1539

F-statistic: 3219 on 5 and 88483 DF, p-value: $< 2.2e-16$

Scatterplots and regression lines were used to visualise the relationship between PM2.5 concentration and hospital admissions across the selected cities.



4 Results

5 Discussion

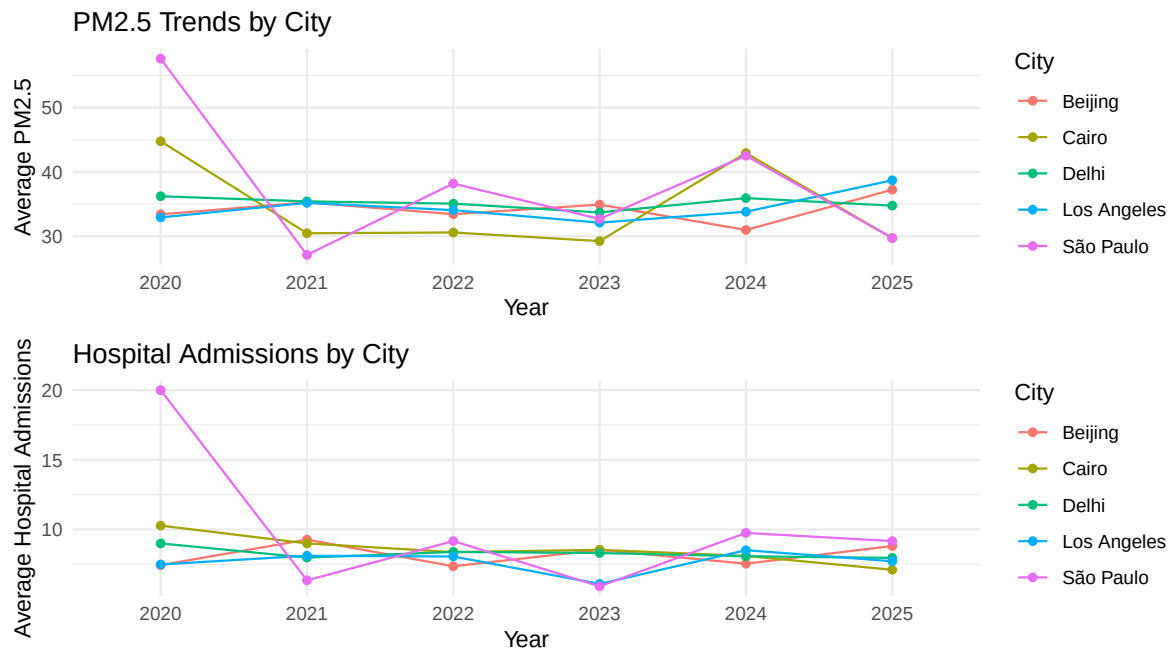
6 Conclusion

7 Recommendations

8 Result

Should be used in results

PM2.5 and Hospital Admissions Trends
Top 5 Most Polluted Cities (2020–2025)



this is an extra filtration we can use in result